

Field Note Journal #6 – What is the Best Driving Route?

The driving plan is as follows to travel through from the Archean age to the Quaternary age:

1. Start at Longlac, Ontario, and go to Nipigon, Ontario via Highway 11. Then, via Highway 17, go straight to Thunder Bay, Ontario (travels through Archean and Proterozoic)
2. From Thunder Bay, Ontario, cross the border with Highway 61 to the United States and go to Duluth, Minnesota, taking the Interstate 35
3. From Duluth, Minnesota, catch US 53 highway passing through Superior, Minnesota, then switch to US 63 highway to Spooner, Wisconsin (One period in Paleozoic (Paibian))
4. From Spooner, Wisconsin, take US 63 to Forest City, Iowa (Givetian (2nd period in Paleozoic))
5. From Spooner, Wisconsin, take US 69 to Omaha, Nebraska, then take US 6 to Lincoln, Nebraska (Mesozoic)
6. From Lincoln, Nebraska, take US-30 to Lexington, Nebraska (Period 1 in Cenozoic (Miocene))
7. From Lincoln, Nebraska, take US-83 to Goodland, Kansas (Holocene)

All of this was from Google Maps and Macrostrat

Field Note Journal #7 – Should I Protect or Produce this Landform?

As a citizen going to buy a home near Toronto, I believe that there are some things to consider within this debate. Aggregates are used to make homes, with concrete being the main use. Also, they are used to making roads, so transportation would be easier with more roads and houses could potentially be cheaper with more of them built and with the

aggregates being local to the area. The only concern that I would have with the expansion would be the destruction of the ecology and ecosystems that are present there, as well as water quality deteriorating due to the dust produced with the aggregates at the quarry. This dust is also harmful for us humans, which I wouldn't personally want to be inhaling.

Sources:

- 1). https://www.ossaga.com/multimedia/0/ossaga_2825_-_infographic_-_who_uses_ontarios_aggregate_final.pdf
- 2). <https://www.torontoenvironment.org/gravel/impacts>
- 3). <https://gravelwatch.org/legacy%20site/laws-gravel.pdf>

Field Note Journal #8 – Is There Concern for Limited Water Resources?

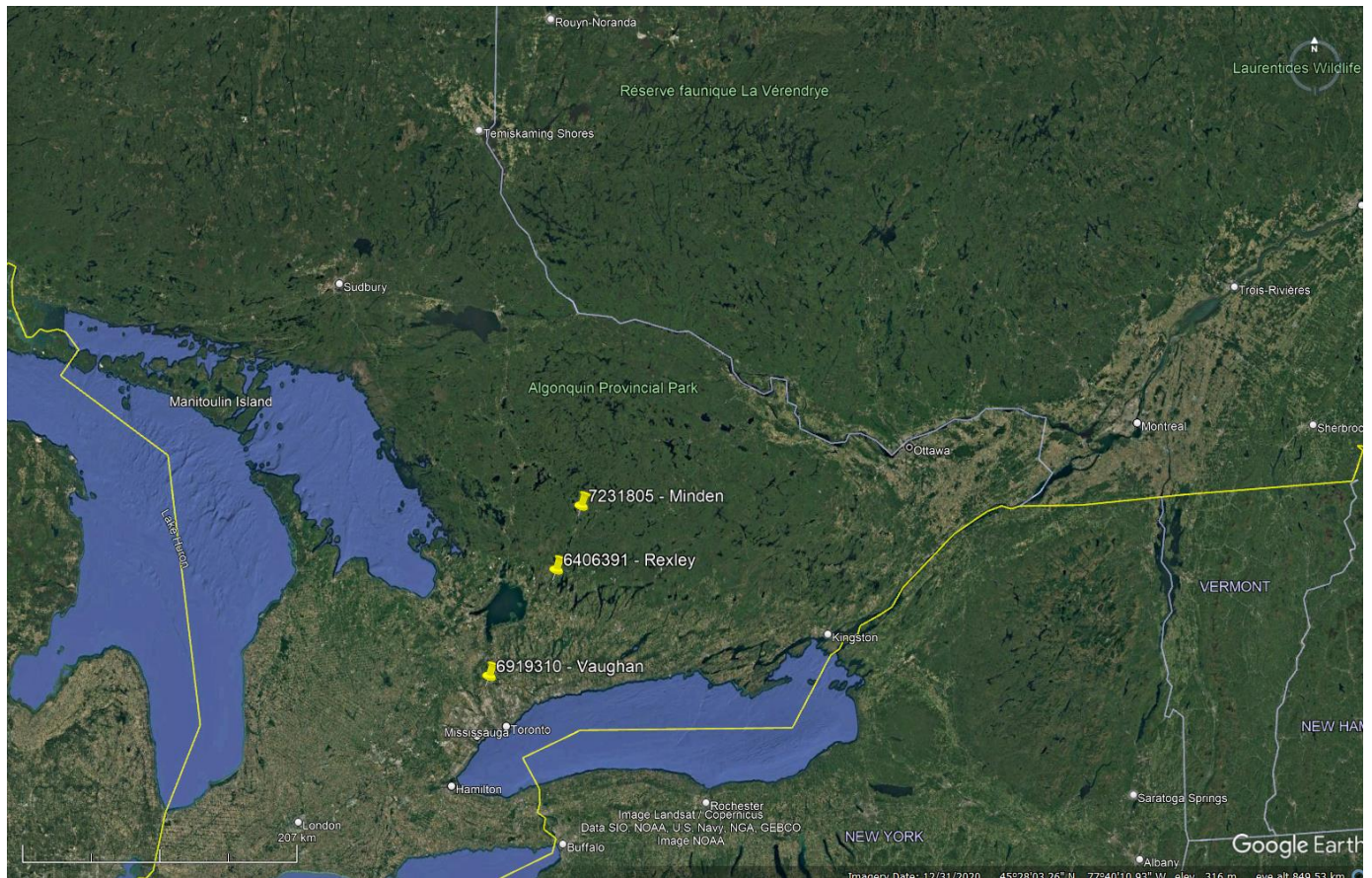
- **Big Idea 9.4** - the human activities that are affecting the availability, quality and distribution of Earth's water are increased population growth, recreational use, and agriculture use
- **Big Idea 7.3** - Water is considered a limited resource in these two examples as it can't be replenished. With ground water, once all that water has been drained, it can't be replenished with more ground water. With surface water, once it dries out or is all used, it can't come back and left in its place will be a drought.
- **Big Idea 5.2** - In each of the examples, ecosystems and humans are negatively impacted, as living organisms need water to survive and thrive on Earth
- **Big Idea 1.1** - Two potential solutions for each of the two examples are as follows:
 - *Ground water* - Increased advocacy of water usage (i.e. using water responsibly for recreational use - showers, cleaning, fixing leaks, etc.)
 - *Surface water* - More efficient agricultural management to grow more food with less land and water through using reservoirs to collect rainwater

Field Note Journal #9 – What Do You See From that Unusual Perspective?

What we see from this unusual perspective of the Great Lakes taken from outer space is how this photograph, when interpreted through the perspective of Earth's internal systems, connects the hydrosphere to three other major Earth systems: the atmosphere, the biosphere, and the geosphere. We see the sun reflecting off of the Great Lakes and clouds made of water covering them, reminding one of the greenhouse effect that the sun and water vapor contribute to in order to keep some energy in and to let most of it out to the solar system, regulating the temperature of the Earth and allowing living organisms to thrive on Earth. As well, seeing the Great Lakes from this view makes one think of the freshwater that is essential to human life as humans need water every day to survive, and how living organisms have come to rely upon water bodies like these to survive, allowing for civilization to continue through the ages and numerous species to propagate. Finally, the way the Great Lakes are shaped and formed reminds one of how water breaks down rocks and how it builds large reservoirs of water like the Great Lakes over thousands of years, shaping the Earth as we know it today.

Field Note Journal #10 – How Deep is Potable Groundwater?

Part 1: The Map



Part 2: The Table of Locations

Water Well Locations	Depth to Water (M)	Lithology	Mapped Lithologic Geology	Bedrock Elevation	Overburden Thickness
7231805	~26	Granite	Crystalline Basement	High	Low
6919310	~44	Med Sand	Shale	Low	High
6406391	~5	Limestone	Limestone	Medium	Low

Part 3: Paragraph Explaining Source of Groundwater

The potential groundwater source for the water well location with granite would be from the fact that the granite could be faulted or when the granite is sheared. This is the reason

why the depth to water is the 2nd largest; granite doesn't store water directly, but only when it's broken. Additionally, the location of the water well is within the Canadian Shield, where the rocks aren't porous. The potential groundwater source for the water well location with limestone would be within the limestone itself, as limestone is quite porous as a material. Therefore, you don't need to drill too deep in this location to find groundwater. The location is in an area where many porous rocks are found, so this also aids in finding groundwater easily. Finally, the potential groundwater source for the location with sand would be quite deep, as shale and sand allow water to drain quite far (very porous), so the top layer is often dry. The reason why groundwater is not found at the same depth/lithology as groundwater follows topography of the land; the fact that human activity can affect the depth of ground water; and lithology isn't completely uniform throughout - different layers have different permeability.